

AURORA SAP.EXE EXECUTION HUNT

Campaign-specific hunt for the Windows encryptor name and compiled command-line flags recovered from an Aurora affiliate.

HUNTING QUERY

CONFIDENCE: MEDIUM

sap.exe

source-recovered Windows variant

-percent -scanners
-noparallel -esxi
-allowfolders

Observable boundary

process name + command line

REPORT DATE

29 August 2026

STATUS

Untested implementation
sketch

PRIMARY TELEMETRY

Defender XDR /
DeviceProcessEvents

ATT&CK

T1486 - Data Encrypted for
Impact

A match is an urgent campaign pivot. It does not prove that encryption completed or that the binary is Aurora without file and incident validation.

CONTROLLED EIGHT-PAGE THREAT DOSSIER



EXECUTIVE ASSESSMENT

Decision-grade summary and analytical boundaries.

ASSESSMENT

OPERATIONAL VALUE

9.2

Fast prevalence check for a source-recovered ransomware encryptor execution pattern.

CLASSIFICATION

Campaign-specific artifact

Hunting query - Untested implementation sketch. Not a durable ransomware detection.

1published query
0production rules
Mediumconfidence
MDEprimary platform

Source fact

CloudSEK recovered Windows sap.exe and Linux/ESXi encrypt.out, both static Zig builds with a shared compiled flag set.

Analytical meaning

The compound process-name and flag signal is highly specific for immediate hunting, but the actor can rename the binary or run defaults without flags.

Assumptions

The observed Windows binary uses at least one selected flag and Defender captures the full process command line.

Promotion boundary

Validate prevalence and benign collision before alerting. Do not block solely on the filename.



THREAT MECHANICS

Source-supported activity and the exact hunting boundary.

BEHAVIOR

01 Domain access

Affiliate reaches interactive or domain-level control.

→

02 Collection

AD data, credentials and policy artifacts are gathered.

→

03 Exfiltration

PowerShell-driven 7-Zip creates 50 GB chunks.

04 Delivery

Encryptors are pulled from Cloudflare R2 and staged.

→

05 Execution

sap.exe exposes compiled tuning and platform flags.

→

06 Impact

Windows anti-recovery precedes encryption.

Condition	Evidence	Query use
sap.exe	Recovered Windows binary	Required
Selected flags	Compiled flag set	At least one
Encryption completion	Campaign behavior	Not inferred
File hash / signer	Not published	Output only

ATT&CK

T1486 - Data Encrypted for Impact. The hunt observes a candidate encryptor process; it does not observe file encryption directly.



TELEMETRY & EVIDENCE MODEL

Minimum data contract and platform limits.

TELEMETRY

REQUIRED

DeviceProcessEvents

Microsoft Defender for Endpoint process creation telemetry with complete command line.

OPTIONAL ENRICHMENT

SHA1, path, parent process, process identity, account, device role, signer lookup and adjacent file/network events.

Field	Reliability	Purpose
Timestamp, DeviceId, DeviceName	Documented	Event and device pivot
FileName, FolderPath	Documented	Executable identity and staging path
ProcessCommandLine	Documented	Compiled-flag filter
ProcessId, ProcessUniqueId	Documented	Process correlation
AccountName, AccountSid	Documented	Execution identity
SHA1, ReportId	Documented	Enrichment and event pivot

Unavailable through this hunt

File-write impact, successful encryption, shadow-copy deletion, System Restore state, binary content and Aurora attribution.

Coverage gap

No command line, renamed executable, default execution or an unonboarded staging host produces a false negative.



DETECTION LOGIC

Direct filters with no threshold, join, sequence or rarity assumption.

KQL

HYPOTHESIS

If the affiliate executes the recovered Windows encryptor with one of its documented flags, process telemetry should show sap.exe plus that flag. Benign overlap is resolved through hash, signer, path, owner and incident context.

KQL - UNTESTED HUNTING

```
DeviceProcessEvents
| where FileName =~ "sap.exe"
| where ProcessCommandLine contains "-percent"
  or ProcessCommandLine contains "-scanners"
  or ProcessCommandLine contains "-noparallel"
  or ProcessCommandLine contains "-esxi"
  or ProcessCommandLine contains "-allowfolders"
| project Timestamp, DeviceId, DeviceName, FileName, FolderPath,
  ProcessCommandLine, SHA1, ProcessId, ProcessUniqueId,
  AccountName, AccountSid, InitiatingProcessFileName,
  InitiatingProcessCommandLine, ReportId
| order by Timestamp desc
```

THRESHOLD	WINDOW	JOIN	KEY	STATUS
None	Retention	None	ProcessUniqueId	Untested

Why Defender XDR

Direct endpoint process telemetry, documented command-line fields and a reliable device/process pivot. No Sentinel schema is assumed.



TRIAGE & INVESTIGATION

Convert a match into defensible incident evidence.

WORKFLOW

1

Confirm process

Verify exact command line, path, time, user and ProcessUniqueld.

2

Validate file

Collect SHA1, signer/certificate metadata, size and binary safely.

3

Pivot backward

Inspect parent, remote session, download source and staging activity.

4

Pivot to impact

Review recovery tampering, file modifications, service and VM activity.

5

Scope enterprise

Hunt hash, path, account, source host and adjacent administrative sessions.

Escalate

Unsigned or unknown binary, server execution, recovery tampering or mass file changes.

Benign

Verified enterprise application named sap.exe with incompatible flags and trusted ownership.

Inconclusive

Missing file, truncated command line or no historical impact telemetry.

Response boundary

Because the candidate is a ransomware encryptor, corroborated matches justify emergency containment. A filename-only collision does not.



FALSE POSITIVES & VALIDATION

What must be proven before alerting.

QUALITY CONTROL

SCENARIO	LIKELIHOOD	DISPOSITION
Enterprise application named sap.exe	M	Compare flags, signer and owner
Test or malware-analysis execution	L	Allowlist lab devices only
Aurora encryptor	L	Emergency investigation
Renamed/default-mode locker	M	Query will miss it

Validation steps

1. Run the hunt across full available retention.
2. Inventory every legitimate sap.exe and its normal command lines.
3. Execute a harmless renamed test binary with one selected flag in a lab.
4. Confirm field population and investigator pivots.

Decision criteria

Success: test execution is visible and production matches are low-volume and dispositionable.

Rejection: a common legitimate application uses the same compound pattern.

Failure: command line or process events are missing.

Allowlisting rule

Allowlist only stable signer/hash/path and verified owner combinations. Never suppress every process named sap.exe.



CONCLUSION & REFERENCES

Coverage statement, platform limits and document control.

FINAL

CONCLUSION

High-value campaign hunt, not a durable production rule.

The exact Windows filename and compiled flag set form a defensible prevalence pivot. Evasion is trivial through rename or default-mode execution, so promotion requires local evidence and broader ransomware coverage.

Cross-SIEM decision

Splunk: no reliable implementation without verified endpoint raw source, index, sourcetype and command-line field extraction.

Chronicle: no reliable YARA-L rule without verified parser and UDM process/command-line mappings.

Deferred ideas

50 GB 7-Zip chunks, generic recovery tampering, ESXi VM termination and Zig rarity. Each lacks exact event-level evidence, verified schema or acceptable benign overlap for this dossier.

References

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3. MITRE ATT&CK - T1486 Data Encrypted for Impact.
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Document control

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